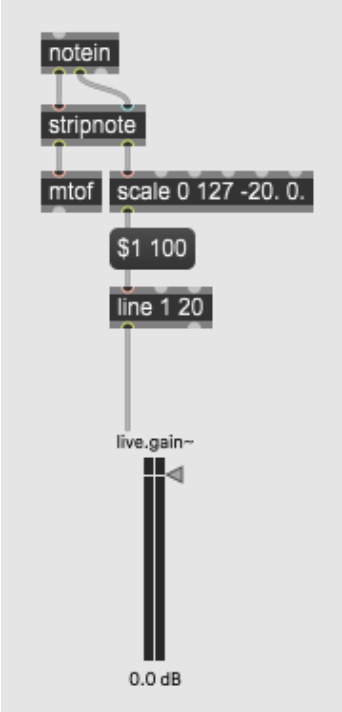
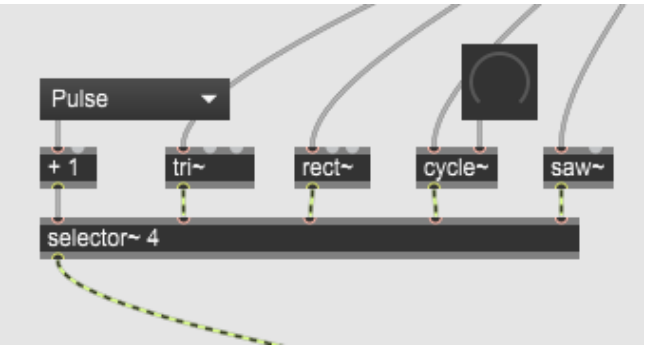
Session 4 — Max4Live & Integration

1. Building a Subtractive Monosynth with Max/MSP

1. Open Ableton Live and navigate to “Max for Live > Max Instrument > Max Instrument” and double click to add a Max Instrument to a MIDI channel.
2. Click the Edit button (3rd button from the right at the top of the Max Instrument object) to open your patch for editing in Max.
3. Press “n” to place a new object. Type **notein** and press enter to place an object that allows notes to come into the patch. Press “n” again and add a new object called **stripnote** that formats incoming notes correctly. Connect notein to stripnote by clicking and dragging from the left outlet on the bottom of the notein to the first inlet at the top of stripnote. Connect the middle outlet of notein to the right inlet of stripnote. Create a new object called **mtof** that converts the incoming note values to frequency in hertz then connect to the first inlet of then stripnote’s right outlet to a **scale** object that scales from 0-127 to -20. to 0. Place a new type of object called a message box with “m” and type the message “$1 100”. Connect the scale object to the message box and connect to a **line** object with an initial value of 1 and grain of 20. Add a **live.gain~** object and connect the line to its first inlet to allow velocity to control its volume. Change the names of live.gain~ to “Gain” in the inspector.

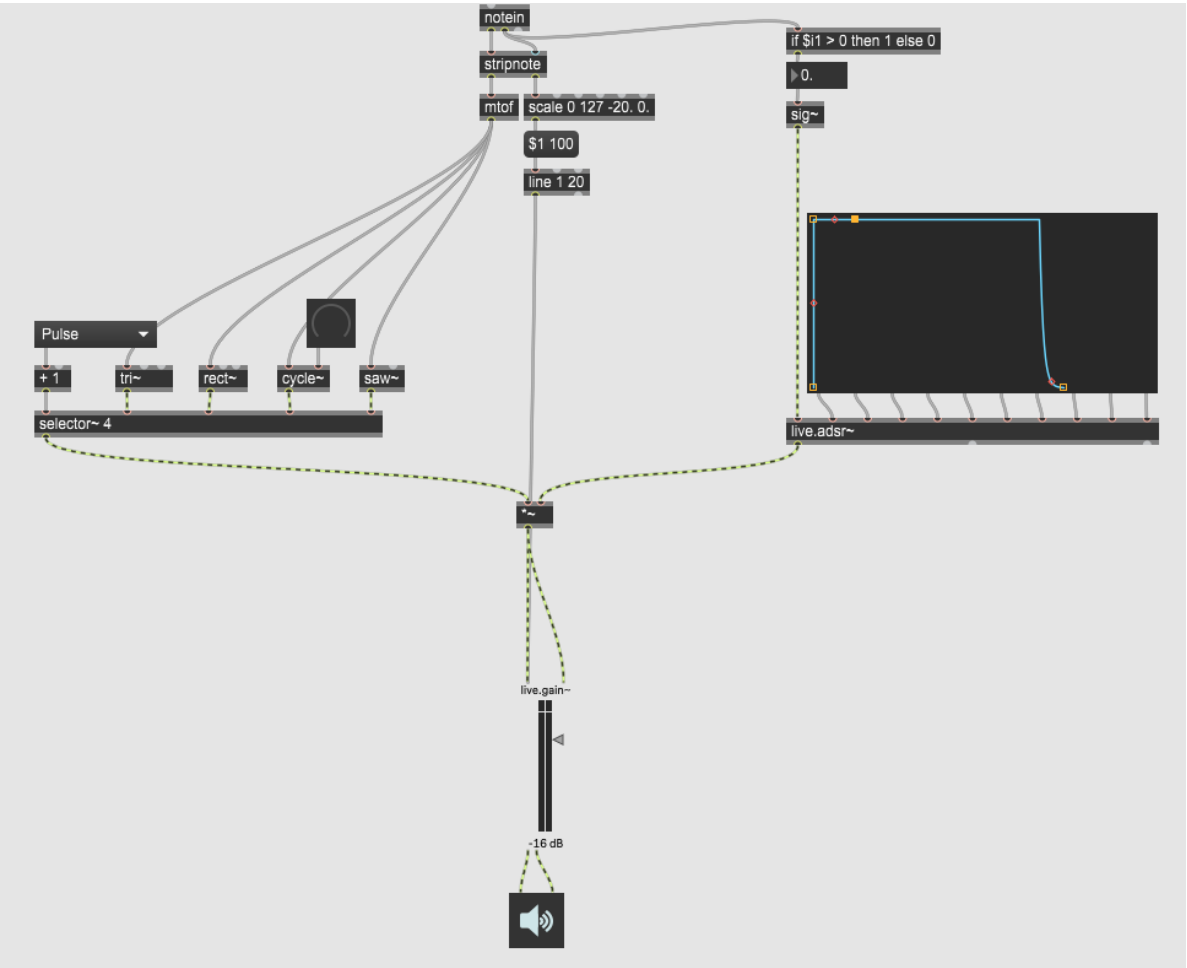


2. Create a **tri~, rect~, cycle~,** and **saw~** object to create a triangle, pulse, sine, and saw wave respectively. Create a **selector~** object with 4 inlets and connect to the latter 4 inlets. In the first inlet, connect a + 1 and a **live.menu**. In live.menu's inspector, name each waveform under Menu Items. Create a **live.dial**. Open the inspector by clicking the *i* icon in the right pane. In the menu on the right, change the scripting name, long name, and short name to "PWM". Change range/enum to 0. 1. and Unit Style to Float. Connect each waveform to mtof.

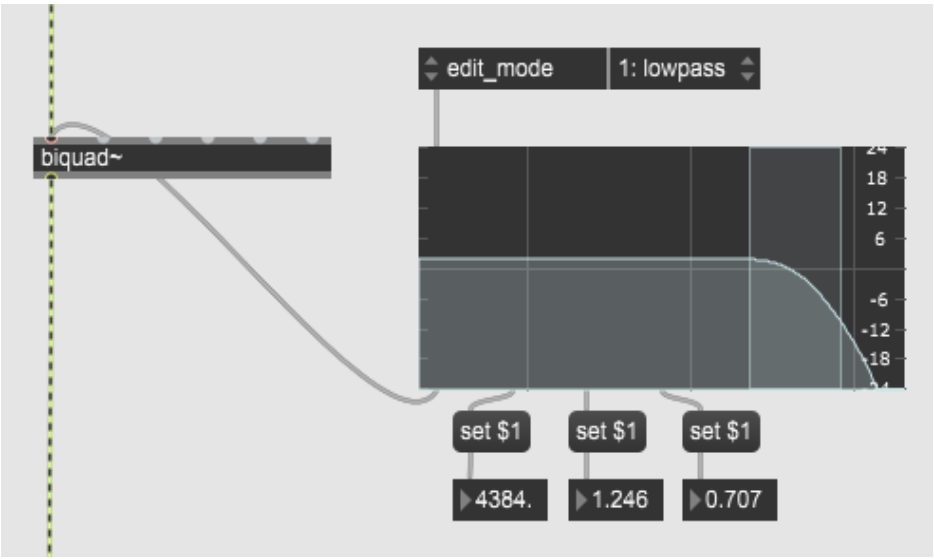


3. To create an ADSR envelope, connect an **if** object to notein's velocity output that reads "if $i1 > 0 then 1 else 0" with a float. Connect this to a **sig~** to convert to a

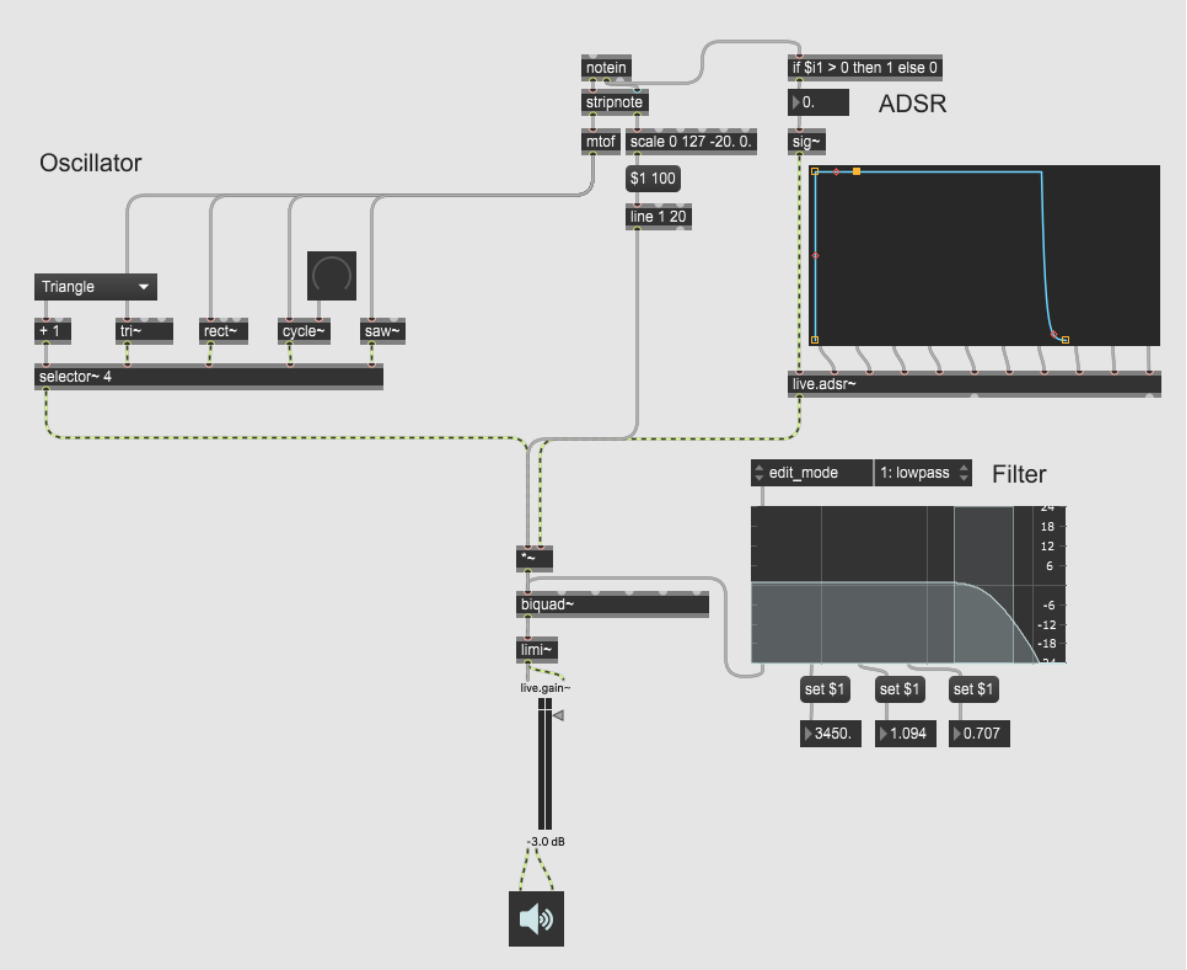
signal. Add a **live.adsr~** and **live.adsrui**. Connect sig~ to live.adsr~'s first inlet and the rest to all of live.adsrui's outlets. Create a *~ object and connect selector~ to the left and live.adsr~ to the right.



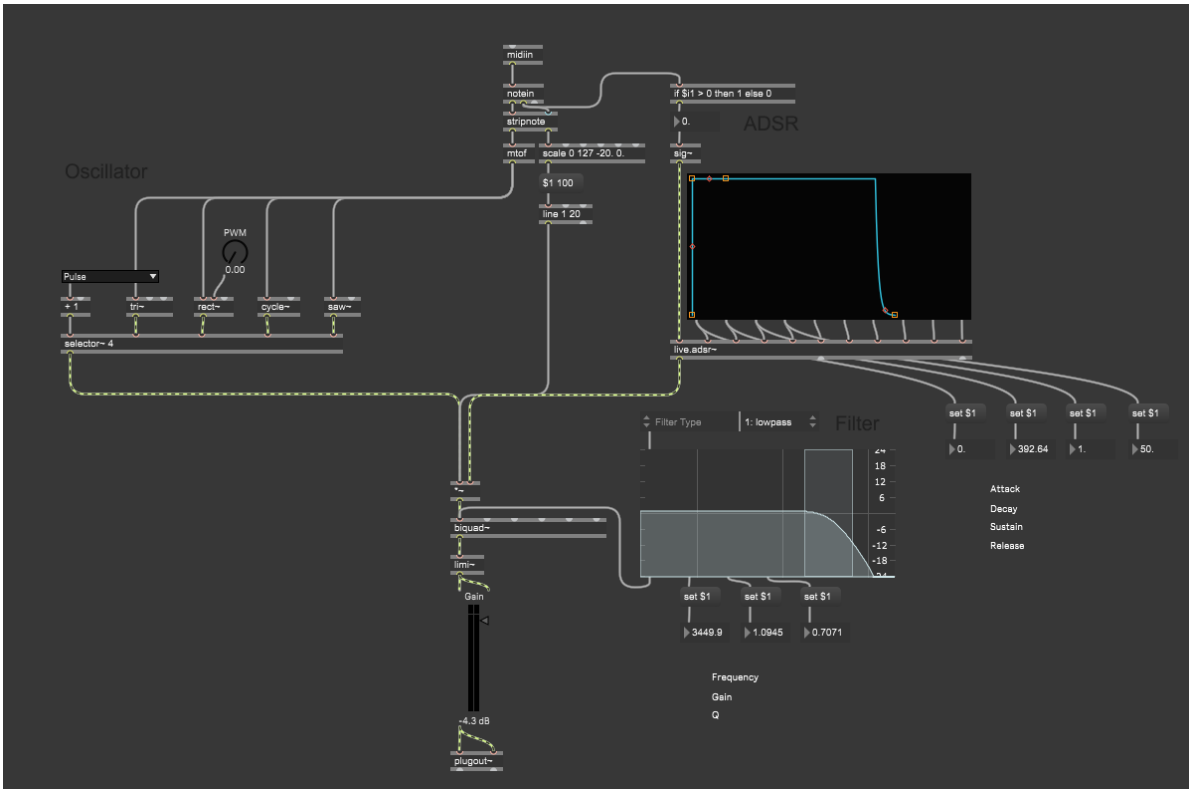
4. Create a **biquad~** filter connected to *~ and a **filtergraph~** to control it. Right click the first inlet of filtergraph~ and select "edit_mode" to change the filter type. Connect the first outlet into biquad~. Create 3 messages containing "set $1" and a float box and connect to the next 3 outlets to see the values of the filter's frequency, gain, and Q respectively. This is your VCF.

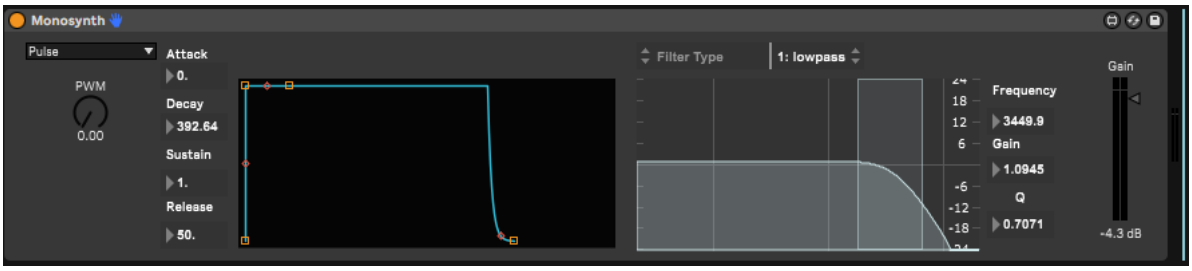


5. Add a **limi~** to create a limiter and connect between biquad~ and live.gain~. Add an ezdac~ and play notes to test your new monosynth!



6. Highlight the oscillator live.menu, live.dial, filtergraph~ along with its 3 corresponding float boxes and edit_mode, live.adsrui, and live.gain~ and add to presentation mode. Open edit_mode in the inspector and change Attribute Display to Label. Check “Show only Object Parameter”.
7. Enter presentation mode and arrange each object to fit within the M4L window.
8. Add comments to annotate as needed.
9. Open the patcher inspector with ctrl+shift+i (or cmd+shift+i), and check “Open in Presentation” under view.
10. Save and close the patch and try playing MIDI notes in Ableton. Try using Live’s native automation feature to automate parameters in the monosynth.


midiin
notein
stripnote
mtof
scale 0 127 -20. 0.
$1 100
line 1 20
if $i1 > 0 then 1 else 0
0.
ADSR
sig~
live.adsr~
Oscillator
PWM
0.00
Pulse
+ 1
tri~
rect~
cycle~
saw~
selector~ 4
Filter Type
1: lowpass
Filter
biquad~
limi~
Gain
-4.3 dB
plugout~
set $1
0.
392.64
1.
50.
Attack
Decay
Sustain
Release
3449.9
1.0945
0.7071
Frequency
Gain
Q


Monosynth
Pulse
PWM
0.00
Attack
0.
Decay
392.64
Sustain
1.
Release
50.
Filter Type
1: lowpass
Frequency
3449.9
Gain
1.0945
Q
0.7071
Gain
-4.3 dB